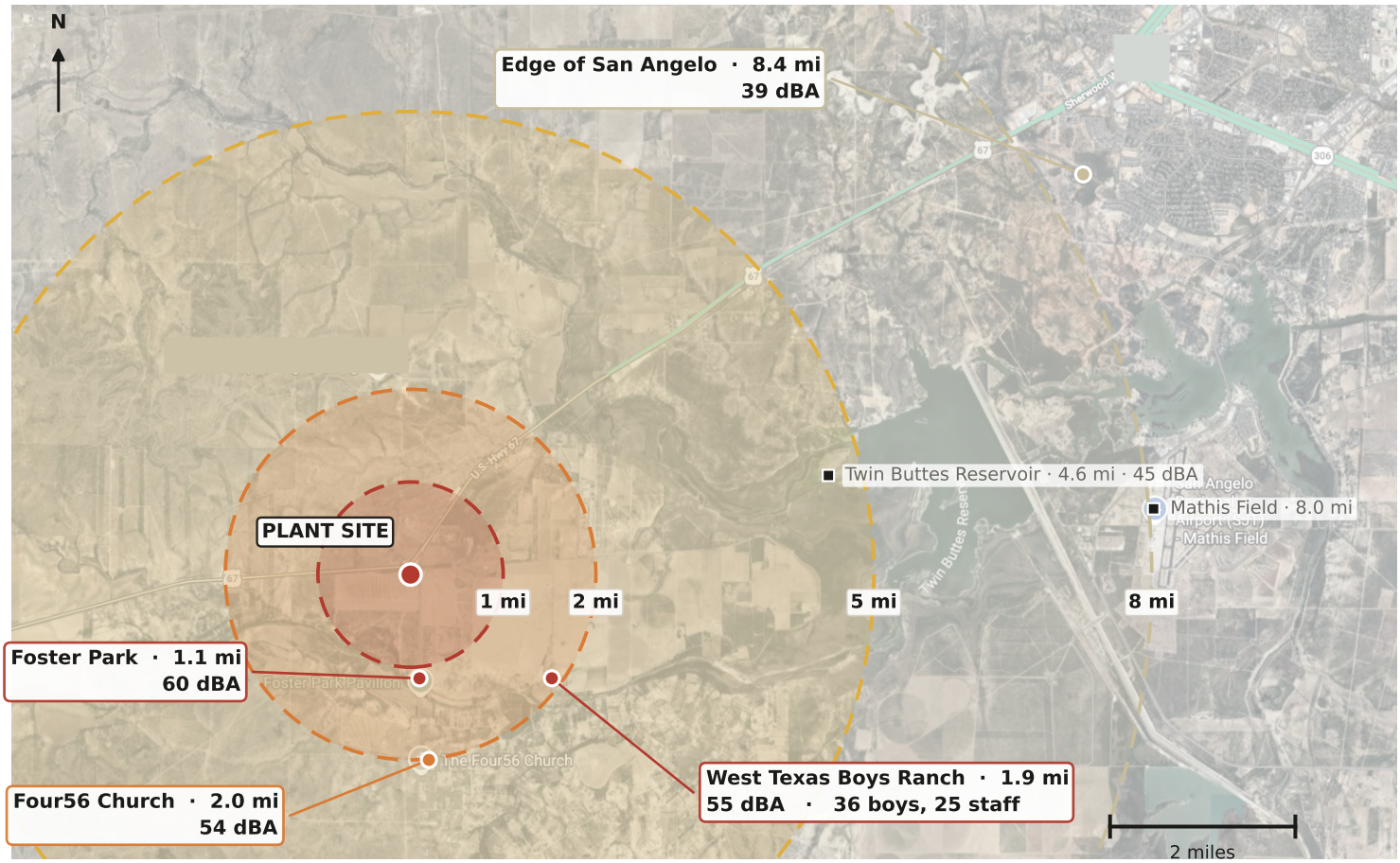
How loud would it be?

A proposed 240-engine gas power plant at US 67 and FM 2335, southwest of San Angelo. TCEQ permit 184380.

At the West Texas Boys Ranch, 1.9 miles away, the calculated level is about 55 dBA — a constant low rumble, day and night. A quiet night there now measures about 29 dBA.

That is roughly five times as loud as the night is now, and it would not stop.



What each ring means

1 mile	**61 dBA**	85 dBC	Like someone talking right beside you
2 miles	**54 dBA**	79 dBC	A steady rumble, day and night
5 miles	**44 dBA**	71 dBC	A clear hum on any still night
8 miles	**39 dBA**	66 dBC	A faint hum, still above natural quiet

dBA is what a sound meter reports. dBC includes the deep rumble that dBA filters out. Both matter; only one gets quoted.

Decibels don't add up the way you'd expect

Going up 10 dB sounds about twice as loud. Doubling the machinery only adds 3 dB. So 240 engines together land about 24 dB above a single one — not 240 times louder, but far louder than any one.

A dBA meter hides the part you feel

dBA filters out bass, because ears are less sensitive to it. dBC keeps it. Here dBC runs 25 to 28 dB higher at every distance. That gap is the rumble — it carries furthest and passes through walls most easily.

Nothing requires anyone to measure it

TCEQ does not weigh noise when it reviews an air permit, and Texas counties have no authority to regulate noise in unincorporated areas. No noise study, no noise data and no noise limit appear anywhere in this application.

SanAngeloDoge.com · **spencer@sanangelodoge.com**

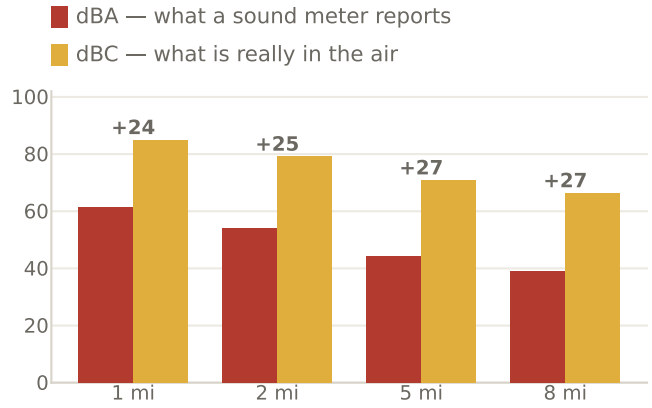
I am a journalist and a San Angelo resident. I have no financial interest in this project or its outcome. Corrections welcome.

Screening estimate, not a measured survey. Method and sources overleaf.

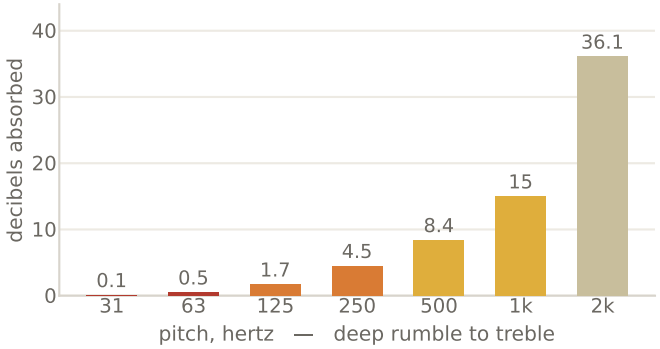
The science, in short

How the number was worked out, what had to be assumed, and what to ask for.

Two things decide what reaches you



The same sound, read two ways. The gap between them is the rumble, and it grows with distance.



How much sound two miles of open air soaks up. Deep sound barely notices. At 4,000 Hz the air removes 117 dB.

Why the rumble is the part that matters

Deep sound travels further and passes through buildings more easily than high sound. A closed-up house cuts high sound by roughly 28 dB but low sound by only about 10 dB. So the further away you are, and the more time you spend indoors, the larger the share of what you notice is rumble — the part a dBA reading leaves out.

How this was worked out

1. Bergen's own factory measurement of this engine family: eight readings at one metre, 103.9 to 113.5 dBA. Averaged by energy, that is 109 dBA.
2. Converted to sound power by the ISO 3744 method. A 43-foot engine is not a point source, and this step alone adds 25 dB a simple distance sum would miss.
3. Twelve engines per hall, minus the building, plus exhaust stacks and radiator fans, times twenty halls.
4. Spread over distance using ISO 9613-2, pitch band by pitch band, including what the air absorbs.

Sense check: that works out to about one millionth of the plant's electrical output leaving as sound, which is normal for heavy industrial machinery.

What had to be assumed

Bergen publishes no sound data for the exact engine model in this application, so figures for a slightly smaller engine in the same family were used. That most likely understates the result rather than overstating it.

Building insulation, exhaust silencers and radiator fans are set at ordinary industry values, because none of them appear in the application. Together they move every number by roughly plus or minus 9 dB.

Levels assume downwind conditions, the standard screening case. Upwind they drop sharply; on a still night they rise. Past about ten miles the weather decides rather than the distance.

What to ask for

- A noise study for the finished plant, all 240 engines running, measured at named places — the Boys Ranch, Foster Park and The Four56 Church.
- Manufacturer sound data for the engine model that is actually being permitted.
- Results given in dBA, dBC and pitch bands, not dBA alone.
- A week of baseline measurements before construction, so any change can be proven afterward.
- A noise limit written into the permit, with a separate limit for low frequency, and verification once running.

Where this has to go

TCEQ does not weigh noise when it reviews an air permit, and Texas counties cannot regulate noise in unincorporated areas. That leaves the Tom Green County Commissioners Court and the State — which is why it is worth putting to them in writing, before the plant is built.

Sources — Engine size, weight and output: Bergen Engines' own fact sheets, bergenengines.com. Sound readings: a Bergen factory measurement sheet filed in a public permit record; preliminary, and not authenticated by Bergen. Method: ISO 3744, ISO 9613-1, ISO 9613-2. Plant description: TCEQ permit application 184380 / PSDTX1710, Westline TX Holdings LLC. Boys Ranch location: wtbr.org. Base map: Google; imagery Airbus, Maxar, Landsat, Copernicus. Full calculations available on request.

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Screening estimate. Figures are calculated, not measured on site.